Fundamentals of Machine Learning

MULTI-CLASS DISEASE DETECTION USING DENSENET AND GRAD-CAM IN CHEST X-RAYS

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **“MULTI-CLASS DISEASE DETECTION USING DENSENET AND GRAD-CAM IN CHEST X-RAYS”** is the bonafide work of **“Prasannakumar (230701238) and Sanjana R (230701285)”**, who carried out the project work under my supervision.

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ABSTRACT

Medical image processing and deep learning techniques are transforming the healthcare industry by enabling faster and more accurate disease diagnosis. Chest X-ray analysis is one of the most important applications of artificial intelligence in the medical field because lung-related diseases such as Pneumonia, Atelectasis, Effusion, Infiltration, Cardiomegaly, and Fibrosis are common and require timely diagnosis. Manual interpretation of chest X-rays by radiologists is time-consuming and may sometimes lead to errors due to workload and human limitations. Therefore, automated disease detection systems are becoming increasingly important.

This project presents a deep learning-based Multi-class Disease Detection System using DenseNet architecture for detecting multiple thoracic diseases from chest X-ray images. The system uses DenseNet-121 as the primary convolutional neural network model because of its strong feature extraction capability and reduced vanishing gradient problem. In addition to disease prediction, the project also integrates Grad-CAM (Gradient-weighted Class Activation Mapping) to provide visual explanations for the model's predictions. Grad-CAM highlights the important regions in the X-ray image that contribute to disease detection, making the system more transparent and interpretable.

The proposed system uses preprocessing techniques, dataset balancing methods, data augmentation, and optimized loss functions to improve performance. The model is trained using the NIH Chest X-ray dataset containing multiple disease classes. Experimental results demonstrate that the system achieves effective classification performance while also generating explainable heatmaps for medical interpretation.

This project helps in reducing diagnostic workload for doctors and provides a reliable computer-aided diagnostic system for healthcare applications.

1. INTRODUCTION

Artificial Intelligence and Deep Learning are rapidly improving the field of medical image analysis. Among all medical imaging techniques, Chest X-rays are one of the most widely used diagnostic tools for detecting thoracic diseases. Chest X-rays are affordable, fast, and easily available in hospitals and healthcare centers. However, interpreting large numbers of X-ray images manually is difficult and requires experienced radiologists.

Deep learning models, especially Convolutional Neural Networks (CNNs), have shown remarkable success in image classification tasks. CNNs automatically learn important features from images without requiring manual feature extraction. DenseNet is one such powerful CNN architecture that improves feature propagation and reduces redundancy by connecting each layer to every other layer.

In this project, a DenseNet-121 architecture is used for multi-class disease detection in chest X-ray images. The model predicts the probability of multiple diseases from a single image. Since deep learning models are often considered "black-box" systems, Grad-CAM is integrated to visualize the regions responsible for predictions. This increases trust and interpretability in medical applications.

The system is designed to classify multiple thoracic diseases efficiently while providing visual evidence through heatmaps. The project combines deep learning, image preprocessing, model optimization, and explainable AI techniques into a single healthcare solution.

Primary Objectives of this Project

1. To develop an automated disease detection system using DenseNet architecture.
2. To classify multiple thoracic diseases from chest X-ray images.
3. To improve diagnostic accuracy using deep learning techniques.

4. To implement image preprocessing and data augmentation techniques.
5. To handle dataset imbalance using weighted loss functions.
6. To provide explainability using Grad-CAM heatmaps.
7. To reduce dependency on manual diagnosis and assist healthcare professionals.
8. To evaluate the model using metrics such as accuracy, precision, recall, F1-score, and AUC.

2. LITERATURE REVIEW

Medical image classification using deep learning has gained significant attention in recent years. Researchers have developed various methods and architectures to improve disease detection from chest X-rays.

Wang et al. introduced the NIH ChestX-ray14 dataset, which contains thousands of chest X-ray images with disease labels. This dataset became one of the most commonly used benchmarks for multi-label thoracic disease classification.

Rajpurkar et al. proposed CheXNet, a 121-layer DenseNet model that achieved radiologist-level performance for pneumonia detection. DenseNet became highly popular because it improves information flow between layers and reduces overfitting.

Selvaraju et al. introduced Grad-CAM, an explainability method that generates localization maps to highlight important regions in images. Grad-CAM helps users understand why a deep learning model predicted a particular class.

Several researchers also explored methods such as ResNet, VGGNet, EfficientNet, Vision Transformers, and U-Net for medical image classification and segmentation. However, DenseNet remains one of the best choices for chest X-ray analysis due to its strong feature reuse and efficient learning capability.

Existing System

Existing systems for chest disease detection mainly rely on manual interpretation by radiologists. Traditional computer-aided diagnosis systems use handcrafted features and classical machine learning algorithms such as Support Vector Machines (SVMs) and Random Forests.

Limitations of existing systems include:

- Lower accuracy due to manual feature extraction.

- Difficulty handling multiple diseases simultaneously.
- Poor generalization on large datasets.
- Lack of explainability in predictions.
- Time-consuming diagnosis process.

Some existing deep learning systems provide only disease classification without explaining the predictions. Therefore, integrating Grad-CAM with DenseNet provides both accurate classification and visual interpretation.

3. PROPOSED SYSTEM

The proposed system is a deep learning-based multi-class disease detection framework using DenseNet-121 and Grad-CAM explainability. The system accepts chest X-ray images as input, preprocesses the images, extracts deep features using DenseNet, predicts disease probabilities, and generates Grad-CAM heatmaps.

The model is trained using chest X-ray datasets containing multiple disease categories. Data preprocessing, augmentation, and class balancing techniques are applied to improve model performance. Grad-CAM is used to highlight important disease regions in the X-ray image.

The proposed system improves classification accuracy, reduces diagnostic errors, and increases transparency through visual explanations.

Key Features

1. **Multi-class disease detection**: The proposed system is capable of detecting multiple thoracic diseases from a single chest X-ray image using a DenseNet-121 deep learning architecture. Instead of identifying only one disease, the model performs multi-label classification and predicts the probability of different diseases such as Pneumonia, Effusion, Atelectasis, Fibrosis, and Cardiomegaly simultaneously. This helps in faster and more efficient medical diagnosis.
2. **DenseNet-121 deep learning architecture**: The system uses DenseNet-121 as the core convolutional neural network model for feature extraction and classification. DenseNet improves information flow between layers through dense connections, which helps reduce the vanishing gradient problem and improves feature reuse. This allows the model to learn complex medical image patterns effectively and achieve higher classification accuracy.

3. **Automatic Disease Probability Prediction**: The system automatically predicts disease probabilities using sigmoid activation functions in the output layer. Each disease is assigned an independent probability score, allowing accurate multi-label classification. This enables doctors to identify the likelihood of multiple diseases from a single image.
4. **Grad-CAM explainability**: The proposed system integrates Grad-CAM (Gradient-weighted Class Activation Mapping) to generate visual heatmaps for model predictions. Grad-CAM highlights the important regions in the chest X-ray image that influenced the disease prediction. This improves interpretability and helps doctors understand the reasoning behind the model's decisions.
5. **Data augmentation support**: Before training, chest X-ray images are preprocessed using resizing, normalization, and augmentation techniques. Data augmentation methods such as image rotation, horizontal flipping, and brightness adjustments are used to increase dataset diversity and improve model generalization. These techniques help the system reduce overfitting and perform better on unseen images.
6. **Weighted loss functions for imbalance handling**: Medical datasets often contain an unequal number of samples for different diseases. To address this issue, the system uses weighted binary cross-entropy loss functions that assign higher importance to minority disease classes. This improves the model's ability to correctly identify rare diseases and increases overall prediction reliability.
7. **Patient-level dataset splitting**: The dataset is divided into training, validation, and testing sets using patient-level splitting techniques. This ensures that X-ray images from the same patient do not appear in multiple datasets, preventing data leakage and improving the fairness of model evaluation.

8. **Threshold Tuning**: Employs an adaptive thresholding mechanism on the validation set to optimize the F1-score for each specific class, compensating for varying disease prevalences.
9. **Visual Output and Heatmap Generation**: Along with disease prediction results, the system generates Grad-CAM heatmap overlays on the original chest X-ray image. These visual outputs help healthcare professionals identify disease-affected regions more clearly and improve confidence in AI-assisted diagnosis.

Dataset Description

The system utilizes the NIH Chest X-ray Dataset, which comprises over 100,000 frontal-view X-ray images with 14 disease labels. The dataset also includes a "No Finding" category for healthy patients. The images are resized to 224x224 pixels and normalized using ImageNet mean and standard deviation values. Data augmentation techniques like random horizontal flips, rotation, and color jittering are applied during training.

Dataset Characteristics

- Large-scale chest X-ray dataset.
- Contains frontal chest X-ray images.
- Includes multiple disease labels.
- Suitable for multi-label classification.

Diseases Included in Dataset

1. Atelectasis
2. Cardiomegaly
3. Effusion
4. Infiltration
5. Mass
6. Nodule
7. Pneumonia

8. Pneumothorax
9. Consolidation
10. Edema
11. Emphysema
12. Fibrosis
13. Pleural Thickening
14. Hernia

Preprocessing Steps

- Image resizing.
- Normalization.
- Data augmentation.
- Dataset balancing.
- Train-validation-test split.

3.1 ARCHITECTURE DIAGRAM

Architecture Explanation

1. Input chest X-ray images are collected from the dataset.
2. Images are resized and normalized during preprocessing.
3. Data augmentation improves generalization.
4. DenseNet-121 extracts deep image features.
5. The classifier predicts disease probabilities.
6. Grad-CAM highlights important disease regions.
7. Final results are displayed with heatmaps.

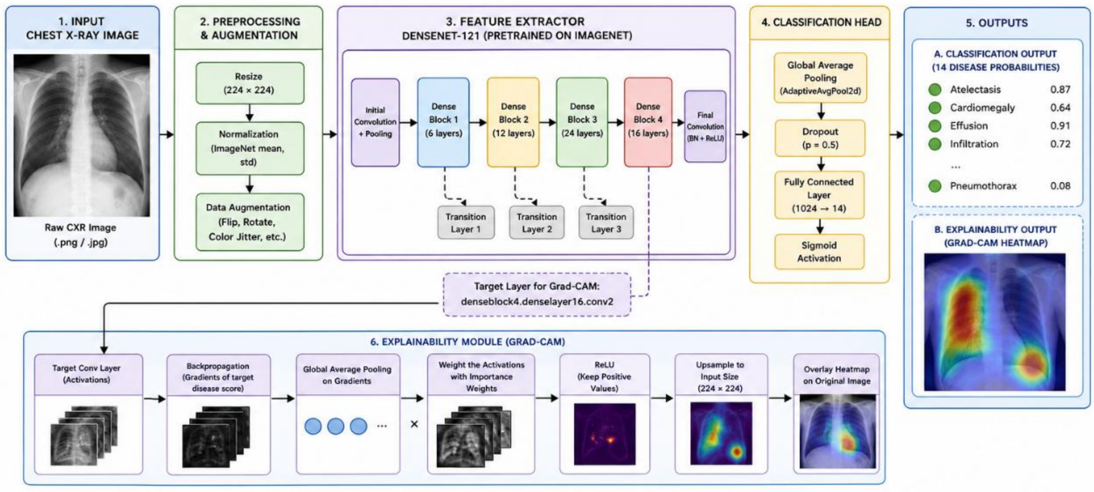

Fig.1: Architecture Diagram

4. MODULES DESCRIPTION

4.1 MODULE 1: Data Pipeline and Model Training

This module is responsible for dataset handling, preprocessing, model creation, and training the DenseNet-based disease detection system. It forms the core learning component of the project and ensures that the model is trained efficiently using optimized deep learning techniques.

Data Pipeline

The data pipeline handles the complete flow of data from dataset loading to preparing batches for model training and validation. The dataset consists of chest X-ray images and corresponding disease labels stored in CSV format.

The module first reads the dataset CSV file and maps image file paths with their associated disease labels. Since the dataset contains a very large number of "No Finding" samples compared to disease-positive samples, an initial undersampling process is performed to reduce class imbalance and improve learning efficiency.

To prevent data leakage and ensure fair evaluation, the dataset is divided into training, validation, and testing sets using unique Patient IDs instead of random image splitting. This guarantees that images from the same patient do not appear in multiple datasets.

Dataset Split

- Training Set – 70%
- Validation Set – 10%
- Testing Set – 20%

Image Preprocessing

Before feeding images into the neural network, several preprocessing operations are applied:

- Image resizing to standard dimensions (224 × 224)
- Image normalization using ImageNet mean and standard deviation
- Conversion into tensor format

Data Augmentation

To improve model generalization and reduce overfitting, multiple augmentation techniques are applied during training:

- Random horizontal flipping
- Random image rotation
- Color jittering
- Random transformations

These augmentations help the model learn robust image features under varying conditions.

DataLoader Configuration

PyTorch DataLoaders are configured for efficient batch processing and faster GPU utilization. The DataLoader handles:

- Batch generation
- Data shuffling
- Parallel image loading
- Efficient memory management

Model Definition

The system uses a custom implementation called CheXNetBaseline, which is built using a pre-trained DenseNet-121 backbone. DenseNet-121 is selected because of its efficient feature propagation and strong performance in medical image classification tasks.

The classifier section of DenseNet is modified to support 14 disease classes. The final classification head includes:

- Adaptive Average Pooling Layer (AdaptiveAvgPool2d)
- Dropout Layer for regularization
- Fully Connected Linear Layer
- Xavier Weight Initialization

The DenseNet architecture enables feature reuse and improves gradient flow between layers, leading to better learning efficiency and reduced vanishing gradient problems.

Training Loop

The training loop performs forward propagation, loss computation, backward propagation, and model optimization.

To improve computational efficiency, the project uses Automatic Mixed Precision (AMP) with:

- GradScaler
- autocast

AMP reduces memory usage and speeds up training while maintaining numerical stability.

Loss Functions

Since medical datasets are highly imbalanced, the system uses:

- WeightedBCEWithLogitsLoss
- Optional FocalLoss

The class weights are calculated inversely proportional to disease frequencies in the training dataset. This ensures that rare diseases receive higher importance during training.

Optimization Techniques

Several optimization strategies are implemented:

- Gradient clipping
- Early stopping
- Learning rate optimization
- Validation monitoring

Early Stopping Mechanism

An EarlyStopping mechanism continuously monitors the validation AUC score during training. If the validation performance stops improving for multiple epochs, training is automatically stopped to prevent overfitting and unnecessary computation.

Advantages of Module 1

- Efficient dataset management
- Prevention of data leakage
- Balanced learning using weighted loss functions

- Faster training with AMP
- Improved generalization using augmentation
- Reduced overfitting through early stopping
- Robust feature extraction using DenseNet-121

4.2 MODULE 2: Evaluation and Explainability (Grad-CAM)

This module evaluates the trained model performance and generates visual explanations for disease predictions using Grad-CAM. It ensures that the model predictions are both accurate and interpretable.

Model Evaluation

The evaluation module processes the testing dataset using the trained DenseNet model and calculates multiple performance metrics.

Performance Metrics

The following metrics are used:

- Accuracy
- Precision
- Recall
- F1-score
- AUC-ROC score

The system calculates per-class AUC-ROC and F1-scores for all 14 disease categories individually.

Threshold Optimization

Instead of using a fixed threshold value such as 0.5 for all diseases, the system dynamically determines the optimal threshold for each disease separately.

The optimal classification threshold is selected by:

- Running predictions on the validation dataset
- Maximizing the F1-score for each disease class

These optimized thresholds are then applied during testing to improve classification accuracy and sensitivity.

ROC Curve Generation

The evaluation module also generates ROC curve plots for all disease classes. These plots help visualize:

- True Positive Rate (TPR)
- False Positive Rate (FPR)
- Classification capability of the model

The ROC curves are automatically saved for analysis and comparison.

Explainability using Grad-CAM

One of the major limitations of deep learning systems in healthcare is the lack of interpretability. To solve this issue, the proposed system integrates Grad-CAM (Gradient-weighted Class Activation Mapping).

Grad-CAM provides visual explanations by highlighting the image regions that contributed most to the prediction.

Working of Grad-CAM

The module loads the best-trained model weights and registers:

- Forward hooks
- Backward hooks

These hooks are attached to the final convolutional layer: denseblock4.denselayer16.conv2

Heatmap Generation Process

1. The input chest X-ray image is passed through the DenseNet model.
2. The gradients of the target disease class are computed with respect to the feature maps.
3. The gradients are globally averaged to obtain neuron importance weights.
4. These weights are multiplied with the corresponding feature maps.
5. A weighted sum of feature maps is computed.
6. ReLU activation is applied to retain only positive contributions.
7. The heatmap is resized to match the original image dimensions.
8. Color mapping is applied using OpenCV colormaps.

9. The heatmap is overlaid on the original chest X-ray image.

Grad-CAM Output

The final output highlights the important disease-related regions in the lungs, helping doctors visually understand why the model predicted a particular disease.

Advantages of Grad-CAM

- Improves model interpretability
- Increases trust in AI predictions
- Highlights disease-affected regions
- Assists radiologists in diagnosis
- Provides explainable AI support

LLM-Based Explainability Integration

In addition to Grad-CAM visualization, the system can be extended with an LLM-based explainability module. This module converts prediction outputs and Grad-CAM observations into human-readable medical explanations.

For example:

"The model identified abnormal opacity regions in the lower lung area, which may indicate signs of Pneumonia."

This combination of visual and textual explainability improves transparency and usability in healthcare environments.

Advantages of Module 2

- Accurate performance evaluation
- Dynamic threshold optimization
- Visual disease localization
- Explainable AI integration
- Improved healthcare reliability
- Better interpretation of predictions

5. IMPLEMENTATION AND RESULTS

The project is implemented using Python and deep learning libraries such as PyTorch and Torchvision. The implementation includes preprocessing, model training, evaluation, and Grad-CAM visualization.

5.1 EXPERIMENTAL SETUP

Hardware Requirements

- Processor: Intel i5/i7 or equivalent.
- RAM: 8 GB or higher.
- GPU: NVIDIA CUDA-enabled GPU.
- Storage: 20 GB free space.

Software Requirements

- Operating System: Windows/Linux.
- Python 3.x.
- PyTorch.
- OpenCV.
- NumPy.
- Pandas.
- Matplotlib.
- Scikit-learn.

Training Configuration

- Model: DenseNet-121.
- Batch Size: 32.
- Epochs: 20–50.
- Optimizer: Adam.
- Learning Rate: 0.001.
- Loss Function: Weighted Binary Cross Entropy.

- Image Size: 224 × 224.

```
Using device: cuda
Dataset composition after undersampling:
  No Finding : 24,144
  Disease    : 51,759
  Total      : 75,903

Split sizes:
  Train : 53,274
  Val   : 7,638
  Test  : 14,991
Found 112120 images across all sub directories
Atelectasis: 5.58
Cardiomegaly: 26.38
Effusion: 4.58
Infiltration: 2.79
Mass: 12.23
Nodule: 11.26
Pneumonia: 53.09
Pneumothorax: 13.76
Consolidation: 15.19
Edema: 31.64
Emphysema: 28.80
Fibrosis: 45.20
Pleural_Thickening: 21.06
Hernia: 316.11
Downloading: "https://download.pytorch.org/models/densenet121-a639ec97.pth" to /root/.cache/torch/hu
b/checkpoints/densenet121-a639ec97.pth
```

Fig.2: Dataset preprocessing and training setup

```
/kaggle/input/datasets/prasannnnaaaaa/nih-dataset/train.py:49: FutureWarning: `torch.cuda.amp.autocas
t(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with autocast():

  Batch [100/833] Loss: 0.5834
  Batch [200/833] Loss: 0.5091
  Batch [300/833] Loss: 0.5433
  Batch [400/833] Loss: 0.5788
  Batch [500/833] Loss: 0.6472
  Batch [600/833] Loss: 0.7044
  Batch [700/833] Loss: 0.5733
  Batch [800/833] Loss: 0.6688

/kaggle/input/datasets/prasannnnaaaaa/nih-dataset/train.py:88: FutureWarning: `torch.cuda.amp.autocas
t(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with autocast():

  Train Loss: 0.6094 | Train AUC: 0.9004
  Val   Loss: 1.9493 | Val   AUC: 0.7980
  Time: 1656.1s | LR: 1.05e-05
Early stopping triggered after 7 epochs without improvement.

Training complete. Best Val AUC: 0.8136
```

Fig.3: Model training showing loss, AUC, and early stopping.

Implementation Steps

1. Load dataset.
2. Preprocess images.

3. Apply augmentation.
4. Split dataset.
5. Train DenseNet model.
6. Validate model.
7. Test performance.
8. Generate Grad-CAM outputs.
9. Display results.

5.2 RESULTS

The DenseNet-based disease detection system achieved good classification performance on chest X-ray images.

Observed Results

1. Accurate detection of multiple thoracic diseases.
2. Improved prediction using weighted loss functions.
3. Better generalization through data augmentation.
4. Reduced overfitting.
5. Effective visualization using Grad-CAM.

Performance Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

AUC Performance Analysis

The Area Under Curve (AUC) metric is used to evaluate the classification capability of the model across training epochs.

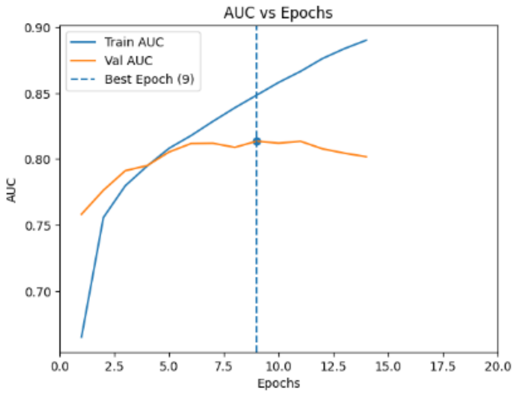

Figure 5.4: Training and validation AUC performance across epochs.

Loss Analysis

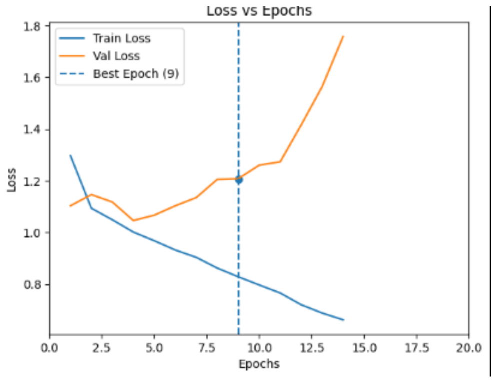

Figure 5.4: Training and validation loss across epochs.

The graph demonstrates that the training loss decreases steadily as the model learns meaningful image features. Validation loss is monitored simultaneously to detect overfitting and evaluate model generalization performance.

Grad-CAM Results

Grad-CAM successfully highlighted disease-affected regions in chest X-ray images. The heatmaps showed that the model focused on medically relevant areas instead of background regions.

Advantages of Results

- Faster diagnosis.
- Improved reliability.
- Explainable predictions.
- Better healthcare assistance.

Sample Output

- Disease prediction probabilities.
- Heatmap overlays.
- Classification results.
- Visualization of affected lung regions.

Multi-class Disease Prediction

```
Model Predictions:
Disease                   Probability   Detected
==================================================
Atelectasis                     0.331      YES ✓
Cardiomegaly                    0.008         no
Effusion                        0.129         no
Infiltration                    0.439      YES ✓
Mass                            0.173         no
Nodule                          0.578      YES ✓
Pneumonia                       0.237         no
Pneumothorax                    0.087         no
Consolidation                   0.133         no
Edema                           0.015         no
Emphysema                       0.243         no
Fibrosis                        0.532      YES ✓
Pleural_Thickening              0.534      YES ✓
Hernia                          0.001         no
```

Figure 5.5: Multi-class disease prediction probabilities.

The output displays probability scores for different thoracic diseases. Diseases with higher confidence values are marked as detected. The model successfully identifies multiple diseases simultaneously, demonstrating effective multi-label classification capability.

Explainability Results using Grad-CAM

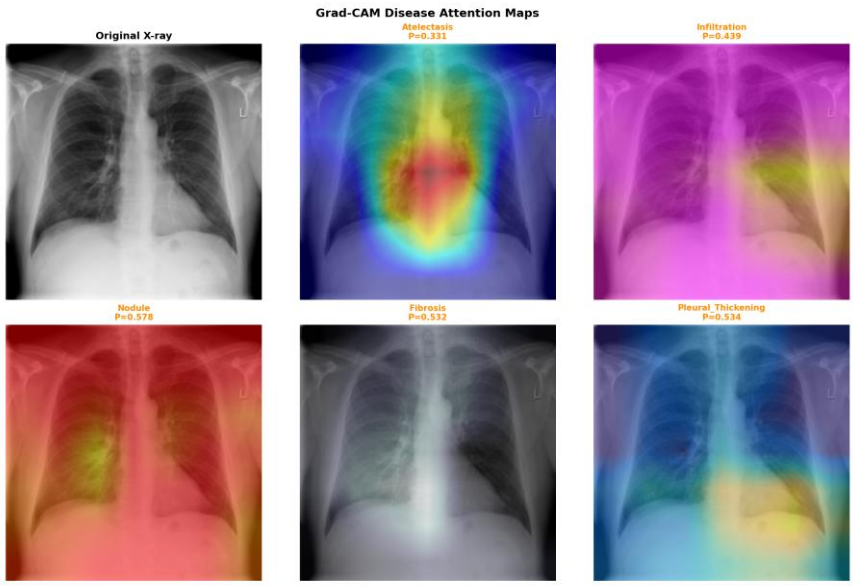

Fig. 5.6: Grad-CAM heatmaps highlighting disease regions.

The Grad-CAM heatmaps visualize the regions responsible for disease predictions such as Atelectasis, Infiltration, Nodule, Fibrosis, and Pleural Thickening. The highlighted areas help explain the model's decision-making process and improve interpretability in medical diagnosis.

6. CONCLUSION AND FUTURE WORK

Conclusion

This project successfully developed a Multi-class Disease Detection System using DenseNet-121 and Grad-CAM for chest X-ray analysis. The system effectively classifies multiple thoracic diseases while also providing explainable heatmaps for visual interpretation.

DenseNet architecture improved feature extraction and classification performance, while Grad-CAM increased transparency by highlighting important disease regions. Data preprocessing, augmentation, and weighted loss functions improved overall model efficiency and reliability.

The proposed system can assist radiologists and healthcare professionals by reducing diagnostic workload and improving disease detection speed.

Future Work

1. Use larger and more diverse medical datasets.
2. Improve accuracy using Vision Transformers.
3. Implement real-time diagnosis systems.
4. Add disease severity prediction.
5. Improve Grad-CAM visualization quality.
6. Deploy the system in hospitals for real-time usage.
7. Use segmentation models for precise disease localization.
8. Integrate LLM-based explainability for generating human-readable medical interpretations and detailed textual explanations of disease predictions.

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